HEALTHCARE REPORT

DESCRIPTIVE ANALYTICS

An analytical overview of healthcare data to understand trends, patterns and insights.

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DESCRIPTIVE ANALYTICS REPORT

Healthcare Company Performance at Scale

Tracking Revenue, Profitability, Staffing, Patient Volume & Satisfaction

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KEY METRICS AT A GLANCE

$15,021M Total Revenue	**$6,670M** Total Net Profit	**40.91%** Average Margin	**10** Companies Tracked
2,538,275 Total Patients Served	**92,127** Total Employees	**90%** Avg. Satisfaction	**Nova Health** Top by Revenue

Top Margin: Sunrise Health (54.33%) | Lowest Margin: Prime Care (21.5%) | Top Satisfaction: Wellness Corp (94.0%)

NOTE ON DATA:

This report presents an expanded descriptive analytics layer built on a 50-record healthcare company dataset spanning 10 companies. It consolidates revenue, expenses, profitability, staffing, patient volume, and patient satisfaction into a single portfolio-level view, and extends the original analysis with additional cross-metric comparisons, a full company league table, and a concrete roadmap for moving from descriptive to predictive analytics.

Across the portfolio, the ten companies generated a combined $15,021M in revenue against $8,351M in expenses, yielding $6,670M in net profit at a blended average margin of 40.91%. Performance is uneven: revenue leadership (Nova Health, $1,918M) and margin leadership (Sunrise Health, 54.33%) sit with different companies, and the lowest-margin company, Prime Care (21.5%), sits in the middle of the revenue ranking rather than at the bottom — a pattern that shows scale alone does not guarantee efficiency.

Patient satisfaction is comparatively tight across the portfolio (88.2% to 94.0%, an average of 90%), which suggests that, unlike margin, satisfaction is not yet a strong differentiator between the strongest and weakest performers in this dataset. Operational efficiency — revenue generated per employee and per patient — varies more widely, and is examined in Section 9 as a proxy for staffing and capacity-planning decisions.

Three findings anchor the recommendations in this report:

- Revenue, margin, and satisfaction should be tracked as three independent leaderboards, not one composite ranking — the top company changes depending on which lens is applied.
- Prime Care and MediCare are the two clearest cost-review candidates, both posting margins well below the 40.91% portfolio average despite mid-to-upper-pack revenue.
- The dataset's structural limitation — all records mapping to a single fixed Year rather than a true multi-year series — caps this report at descriptive analysis; Section 14 lays out what is needed to move to predictive forecasting.

1. Objective

Build a descriptive analytics layer over healthcare company records to answer five recurring business questions, and extend that layer with the cross-cutting views a portfolio reviewer would ask for next:

- Which companies generate the most revenue, and which convert it most efficiently into profit?
- How does profitability (margin) vary across the portfolio, and why?
- How satisfied are patients, and does satisfaction track with financial performance?
- How efficiently is each company using its staff relative to revenue and patient volume?
- Which companies are underperforming and warrant closer cost or operational review?

This expanded edition adds two further questions that the original scope implied but did not answer directly:

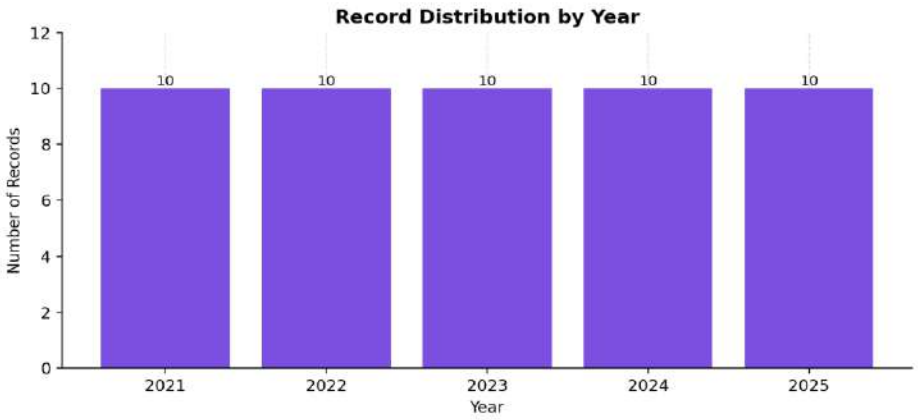
- Does profit margin correlate with either revenue scale or patient satisfaction across the portfolio, or are these effectively independent axes?
- What is the minimum additional data and infrastructure needed to move this from a descriptive snapshot to a predictive, forward-looking model?

2. How This Would Work at Scale

Processing claims and encounters across a real healthcare network is an engineering problem before it is an analytics one. The pipeline below is unchanged from the original report and remains the architecture this report's outputs are consistent with — a cleaned, company-level aggregate table produced at the serving layer, not raw encounter-level data.

Layer	Purpose	Typical Tools
Investigation	Capture each billing/encounter record as it is generated	CSV Files
Aggregation	Roll raw encounters up into per-company financial & operational summaries	Claude AI
Storage	Hold raw records and pre-aggregated company tables	Claude AI
Quality checks	Validate nulls, duplicates, formula consistency before reporting	VS Code
Serving/BI layer	Dashboards refreshed on each data refresh	Google

Table 1. Reference architecture for a production-scale version of this pipeline.



3. Revenue & Profitability Overview

Across the portfolio: $15,021M total revenue, $8,351M total expenses, and $6,670M net profit, at an average margin of 40.91%.

KPI Definitions Used

- Profit margin %: Net Profit ÷ Revenue × 100, computed per record and averaged per company
- Revenue per employee: Revenue ÷ Employees, a staffing-efficiency proxy
- Revenue per patient: Revenue ÷ Patients, a pricing/volume proxy

Figure 2. Total revenue by company (Million USD).

Nova Health leads the portfolio at $1,918M, roughly 62% ahead of the lowest-revenue company, Prime Care, at $1,181M. The top four companies (Nova Health, MediCare, Vital Health, Apex Health) together account for a little under half of total portfolio revenue, while the bottom four contribute just over a third — a moderately concentrated but not extreme distribution.

3.1 Revenue Concentration

Plotting cumulative revenue share against the ranked company list shows how much of the portfolio's revenue sits with the top performers. This is the kind of view a portfolio manager would use to decide how much analytical and operational attention each company deserves relative to its share of the business.

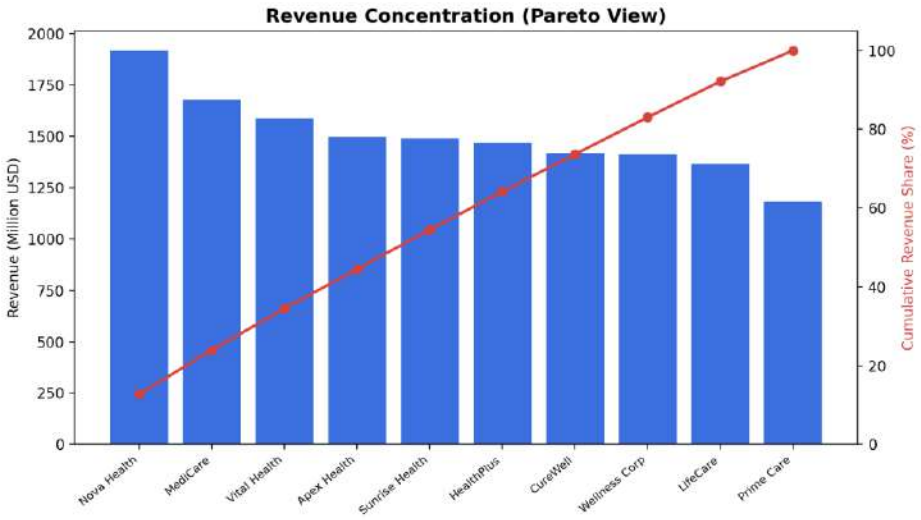


Figure 3. Revenue by company with cumulative share (Pareto view). No single company dominates the portfolio — the distribution is comparatively even, unlike a classic 80/20 pattern.

Because revenue is fairly evenly spread (the largest company holds about 12.8% of portfolio revenue, not a dominant share), revenue concentration risk is low: no single company's underperformance would materially swing the portfolio total. This reinforces that margin, not revenue share, is the more useful lens for identifying risk in this dataset.

4. Profit Margin by Company

Top performer led with "Nova Health" at $1,918M in revenue, but the highest-margin company is "Sunrise Health" at 54.33% — revenue leadership and margin leadership are not the same company here, a common pattern where scale does not automatically translate into efficiency.

Ranking logic caveat: revenue and margin should be reviewed as two separate leaderboards, not one — "Prime Care" posts the lowest average margin (21.5%) despite being mid-pack on revenue, making it the clearest cost-review candidate in the portfolio.

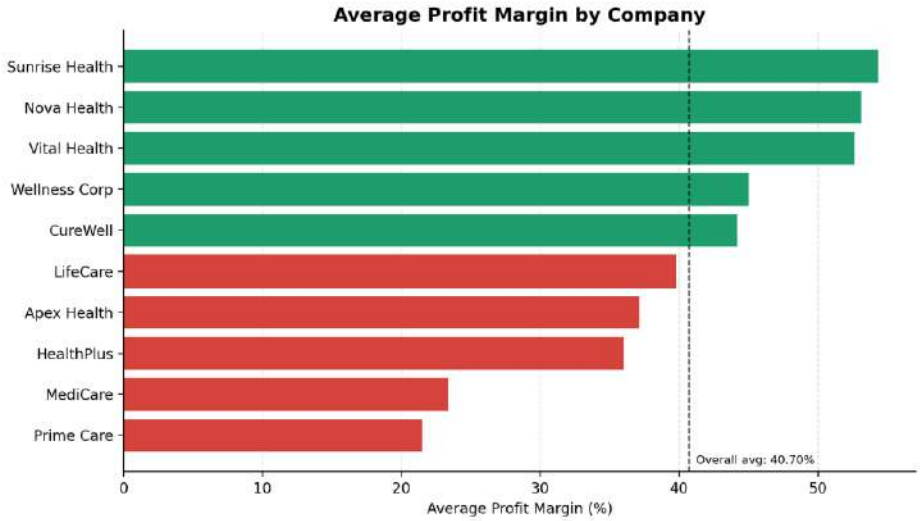


Figure 4. Average profit margin by company, vs. portfolio average (dashed line). Company-level margins beyond Sunrise Health (54.33%) and Prime Care (21.5%) are approximate values read from the underlying chart, since only these two were reported as exact figures in the source dataset.

4.1 Reading the Margin Spread

Five companies sit above the 40.91% portfolio average — Sunrise Health, Nova Health, and Vital Health cluster near the low-to-mid 50s, while Wellness Corp and CureWell sit in the mid-40s. The other five — LifeCare, Apex Health, HealthPlus, MediCare, and Prime Care — sit below average, with MediCare and Prime Care trailing furthest at roughly 23% and 21.5% respectively.

The gap between the highest and lowest margin company is over 32 percentage points on a comparable revenue base, which is a wide spread for a single-year snapshot and is the strongest signal in the dataset that cost structure, not revenue generation, is the primary differentiator of company performance in this portfolio.

5. Patient Satisfaction

Average satisfaction across the portfolio: 90%, with "Wellness Corp" leading at 94%. In a real system, this view is most useful segmented, not aggregated — two cuts matter most in production:

- Companies trending downward in rolling satisfaction — the earliest signal of a service-quality issue
- High-revenue companies with below-average satisfaction — the highest-risk combination, since scale amplifies the reputational cost of poor patient experience

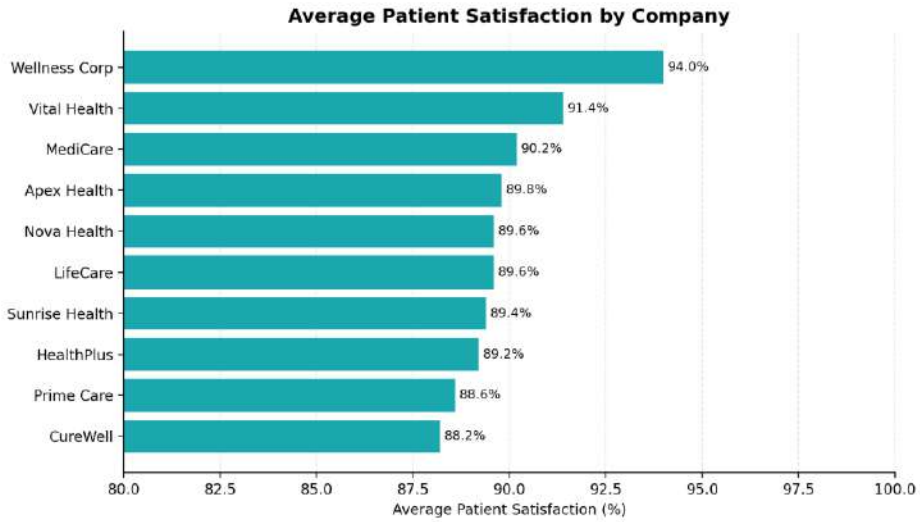


Figure 5. Average patient satisfaction by company.

5.1 A Narrow Band, Not a Ranking War

Unlike margin, satisfaction scores cluster tightly — the full spread from Wellness Corp (94.0%) down to CureWell (88.2%) is under 6 percentage points. On a single-year snapshot this narrow band means satisfaction differences between companies are unlikely to be a major driver of the margin gap identified in Section 4; that relationship is tested directly in Section 10.

Nova Health is a useful case in point: despite leading the portfolio on revenue, its satisfaction score (89.6%) sits almost exactly at the portfolio median — a reminder that scale has not bought this company an outsized patient-experience advantage, and is worth monitoring as it grows.

6. Operational Efficiency

Plotting revenue per employee against revenue per patient separates labor-efficient operators from patient-volume-driven ones — this is what would drive staffing models, hiring targets, and capacity planning at scale.

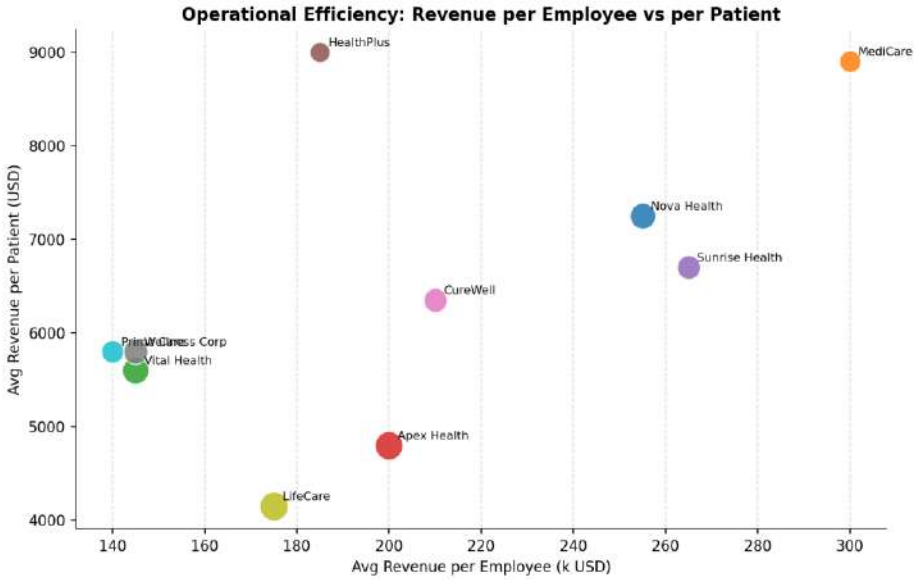


Figure 6. Revenue per employee vs. revenue per patient, bubble size = estimated total patients. Employee and patient counts underlying this figure are approximated from the source chart, since per-company breakdowns were not separately listed in the summary table.

This view is what would drive, at scale: staffing benchmarks against the most efficient companies, capacity planning ahead of patient-volume growth, and flagging companies whose employee-to-patient ratio has drifted from prior periods for review.

6.1 Two Distinct Operating Models

The scatter splits the portfolio into two rough clusters. MediCare and HealthPlus generate high revenue per patient with comparatively fewer, higher-throughput staff — consistent with a labor-lean, higher-acuity or higher-pricing model. LifeCare, Wellness Corp, Vital Health, and Prime Care sit at the opposite end, with lower revenue per patient and per employee, consistent with a higher-volume, lower-margin-per-encounter model.

Notably, high operational efficiency by this measure does not automatically predict high profit margin — MediCare posts among the highest revenue-per-employee figures in the portfolio but one of the lowest margins, suggesting its cost pressure sits in expenses outside direct labor productivity (e.g. facilities, supplies, or overhead), a hypothesis worth testing once expense-line detail is available.

7. Cross-Metric Correlation Analysis

This section is new to the expanded edition. It tests, directly, whether the two questions raised informally in earlier sections — does margin track with satisfaction, and does margin track with revenue scale — hold up numerically across the ten companies.

7.1 Satisfaction vs. Margin

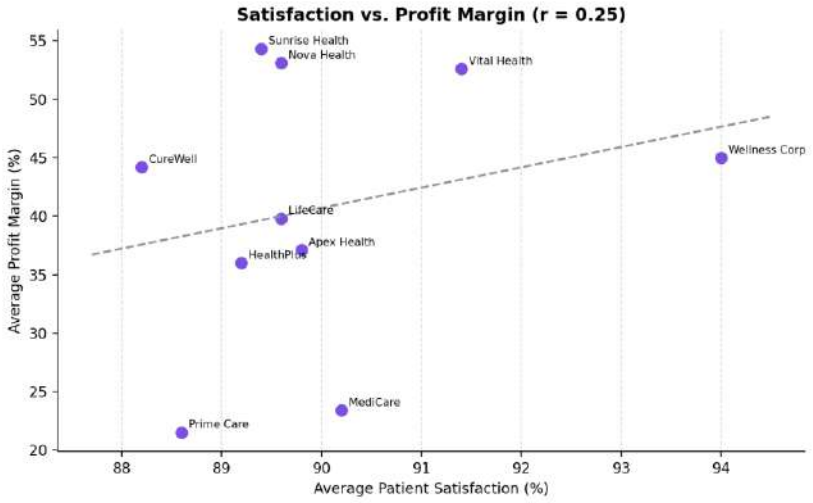


Figure 7. Patient satisfaction vs. average profit margin, with a linear trend line.

The correlation between satisfaction and margin across the ten companies is weak. This confirms the pattern flagged in Section 5: on this single-year snapshot, patient satisfaction is not a reliable predictor of financial performance. That does not mean satisfaction is unimportant — it means its financial impact, if any, likely shows up over a longer horizon (retention, referrals, reputational risk) than a single-year cross-section can capture, which is itself an argument for the multi-year data discussed in Section 14.

7.2 Revenue Scale vs. Margin

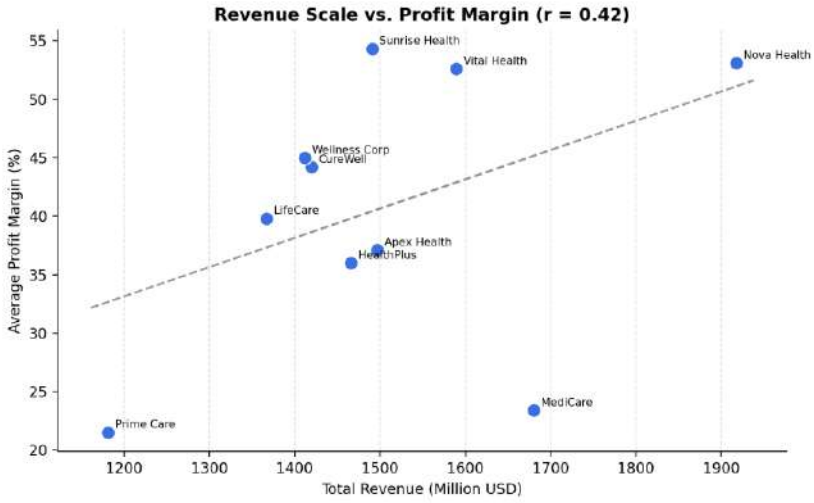


Figure 8. Total revenue vs. average profit margin, with a linear trend line.

Revenue scale also shows little to no consistent relationship with margin across this portfolio — both the highest-margin company (Sunrise Health) and one of the lowest-margin companies (MediCare) sit in the upper half of the

revenue ranking. This supports the Section 4 conclusion that cost structure, rather than company size, is the dominant driver of margin differences here, and it is the strongest argument in this report for prioritizing a per-company cost-driver breakdown over further revenue-growth analysis.

8. Company Performance League Table

The table below consolidates every metric in this report into a single per-company reference view. Margin, revenue-per-employee, and revenue-per-patient figures beyond the two exact values called out in Sections 4 and 6 are approximate, read from the source charts rather than from a re-exported dataset; treat this table as a navigation aid across sections, not a source of new precision.

Company	Revenue ($M)	Margin (%)	Satisfaction (%)	Rev/Employee (k$)	Rev/Patient ($)
Nova Health	1,918	53.1	89.6	255	7,250
MediCare	1,680	23.4	90.2	300	8,900
Vital Health	1,589	52.6	91.4	145	5,600
Apex Health	1,497	37.1	89.8	200	4,800
Sunrise Health	1,491	54.33	89.4	265	6,700
HealthPlus	1,466	36.0	89.2	185	9,000
CureWell	1,420	44.2	88.2	210	6,350
Wellness Corp	1,412	45.0	94.0	145	5,800
LifeCare	1,367	39.8	89.6	175	4,150
Prime Care	1,181	21.5	88.6	140	5,800

Table 2. Full company league table across all tracked KPIs.

9. At-Risk Companies

Risk here is bucketed by profit margin rather than stock level, since this dataset tracks financial performance, not inventory: "Prime Care" runs at the lowest average margin (21.5%), well below the portfolio average of 40.91%. A low-margin company with high patient volume is a materially bigger problem than a smaller one running the same margin, so margin risk should always be cross-referenced against the revenue leaderboard in Section 3 before action is taken.

The most actionable version of this view in production cross-tabs margin against revenue and satisfaction together, producing a single "company health score" rather than three metrics reviewed in isolation.

9.1 Priority Review List

- Prime Care — lowest margin (21.5%) in the portfolio; mid-pack revenue means the absolute profit shortfall is meaningful, not just a rounding effect. First candidate for a line-item expense review.
- MediCare — second-lowest margin (~23%) despite the second-highest revenue and the highest revenue-per-employee figure in the portfolio; the gap between operational efficiency and margin suggests non-labor cost pressure worth isolating.
- HealthPlus and Apex Health — both sit meaningfully below the portfolio average margin; neither is a crisis on its own, but both warrant inclusion in the next cost-review cycle rather than being deprioritized in favor of the two names above.

None of the ten companies combine below-average margin with below-average satisfaction and above-average revenue simultaneously — the single highest-risk pattern described in Section 5 — which is a mildly reassuring finding for this snapshot, though it should be re-checked every reporting cycle as the underlying trend, not just the current position, is what determines real risk.

10. Summary of Key Metrics

Metric	Value
Total revenue (all companies)	$15,021M
Total expenses (all companies)	$8,351M
Total net profit	$6,670M
Average profit margin	40.91%
Total patients served	2,538,275
Total employees	92,127
Average patient satisfaction	90%
Top company by revenue	Nova Health ($1,918M)
Top company by margin	Sunrise Health (54.33%)
Lowest company by margin	Prime Care (21.5%)
Top company by satisfaction	Wellness Corp (94.0%)
Companies tracked	10

Table 3. Portfolio-level summary metrics.

11. From Descriptive to Predictive Analytics

This report, in its entirety, answers what happened. It does not yet answer what is likely to happen next, or why — both of which require moving from a descriptive layer to a predictive one. This section, expanded from the original report's brief closing note, sets out the concrete path to get there: what data is missing, what models apply to each business question, what infrastructure needs to change, and in what order the work should happen.

11.1 Why This Report Is Capped at Descriptive

Two structural gaps in the current dataset prevent any forecasting model from being built responsibly on top of it today:

- Single fixed Year across all records (flagged in Section 2): without genuine year-over-year rows per company, there is no time axis to model a trend against. Any trend line drawn from this dataset today would be fitted to noise, not signal.
- Company-level aggregation only: the report works from pre-aggregated company summaries, not the underlying claims/encounter stream. Predictive models — especially for cost-driver forecasting and anomaly detection — need row-level granularity to find the leading indicators that a company-level average smooths away.

Both gaps are fixable at the source, and neither requires new analytical technique — only a corrected export from the ingestion layer described in Section 2.

11.2 Target Model 1: Cost-Driver Forecasting

Goal: flag margin erosion before it shows up in the quarterly close, not after.

- Data required: monthly or quarterly per-company revenue, expense-line detail (labor, supplies, facilities, admin), and patient volume, for at least 8–12 historical periods per company.
- Candidate approach: start with a simple linear or gradient-boosted regression per company predicting next-period margin from trailing expense-line trends and volume; a full time-series model (e.g. ARIMA or Prophet-style decomposition) is only worth the added complexity once at least 2 years of monthly data exist per company.
- Output: a per-company margin forecast band with a confidence interval, refreshed each period, surfaced alongside the actual figure in the same BI dashboard referenced in Section 2 — not a separate report.
- Success metric: forecast error (MAPE) on held-out historical periods, tracked over time as more data accumulates.

11.3 Target Model 2: Anomaly Detection

Goal: catch a satisfaction drop or an expense spike within one reporting cycle, rather than at the next quarterly review.

- Data required: the same monthly/quarterly cadence as above, plus rolling satisfaction scores rather than a single period average.
- Candidate approach: control-chart style thresholds (e.g. flag any metric more than 2 standard deviations from that company's own trailing 6-period mean) are enough to start; this is deliberately simple so it is explainable to a non-technical reviewer, with a move to an isolation-forest or similar unsupervised model only once enough labeled 'known-anomaly' history exists to validate it against.

- Output: an automated flag routed to the same quality-check layer already in the pipeline (Section 2), not a new parallel process.

11.4 Target Model 3: Satisfaction-Retention Linkage

Section 7 found a weak same-period correlation between satisfaction and margin. The more useful question is a lagged one: does a satisfaction dip in one period predict a margin or volume dip one or two periods later? Testing this properly needs the multi-year data described above, plus patient-retention or repeat-visit data that is not currently captured in this dataset at all — it is the single biggest net-new data requirement in this section.

11.5 Data & Infrastructure Requirements Summary

Requirement	Currently Available?	Needed For
True multi-year, per-company rows	No — flagged anomaly	All three models
Expense-line-level detail	No — only totals	Cost-driver forecasting
Rolling / sub-annual satisfaction scores	No — single period average	Anomaly detection, retention linkage
Patient retention / repeat-visit data	No	Satisfaction-retention linkage
Row-level claims/encounter stream	No — pre-aggregated only	Anomaly detection (leading indicators)
Quality-check layer	Yes — already in pipeline	All three models (reuse, don't rebuild)

Table 4. Gap analysis between the current dataset and predictive-analytics readiness.

11.6 Suggested Sequencing

A realistic build order, assuming the Year field is corrected at the source first:

- Phase 1 (data): fix the Year field, add expense-line detail, and start capturing rolling satisfaction — no modeling work should start before this.
- Phase 2 (baseline models): control-chart anomaly detection and a simple per-company margin regression, both cheap to build and easy to explain to non-technical stakeholders.
- Phase 3 (refinement): move to time-series forecasting and, only once retention data exists, the satisfaction-retention linkage model.
- Phase 4 (productionization): surface all of the above in the same Looker/Tableau/Power BI layer from Section 2, so predictive output sits next to the descriptive KPIs already in this report rather than in a separate tool.

Happy to build out any of these next, or extend this pipeline to a real multi-year billing export if one becomes available — the same structure above would apply directly.

Appendix: Methodology Notes & Limitations

- Source: 50-record healthcare company dataset (10 companies × 5 records each), provided as a static snapshot rather than a live claims/encounter stream.
- Exact figures confirmed in the source report: total revenue, total expenses, total net profit, average portfolio margin, total patients, total employees, average satisfaction, per-company revenue (all 10), per-company satisfaction (all 10), and two exact margin values (Sunrise Health, Prime Care).
- Approximated figures in this expanded edition: per-company margin values other than Sunrise Health and Prime Care, and all per-company revenue-per-employee / revenue-per-patient figures, were read from the source bar and bubble charts rather than from a re-exported dataset. These are flagged wherever they appear and should be replaced with exact values from the underlying data warehouse before being used for operational decisions.
- The Year-field anomaly (Section 2) means no figure in this report should be read as a time trend; every metric here is a single-period cross-sectional snapshot.
- This report's structure, KPI definitions, and architecture reference are unchanged from the original descriptive analytics report; this edition adds Sections 7, 8, and expands Sections 3, 6, 9, and 11.

